Isaac **Pedisich**

Software Engineer

Philadelphia, Pennsylvania

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Lead software engineer with experience in academia and industry. Research experience across both Neuroscience and Computer Science. Sixteen years of professional experience in scientific application development. Passionate about speed, efficiency, and automation. Seeking opportunities to solve hard problems in elegant ways and to apply my skills to cutting-edge research in Alzheimer's Disease.

Technical **Skills**

Expertise in: Python (numpy, pandas, scikit-learn, matplotlib), MATLAB, SQL, Bash, Linux, Git, LaTeX, Go, C, C++

Experience with: Javascript (React, Node.js, D3, JQuery), Java, Rust, MongoDB, Django, Spark, PyTorch, PHP

Professional **Experience**

Pipe Technologies — *Remote*

Manager and Lead Engineer - Internal Systems — *2022 - PRESENT*

- Managed team of eight engineers, setting priorities and engaging with stakeholders to ensure team efficacy
- Maintained and monitored databases, performing zero-downtime migration of 10+TB PostgreSQL services
- Designed and managed data pipelines, building data warehouse from databases, external APIs, and partner-controlled cloud storage
- Architected kubernetes clusters, allowing seamless deployments for multiple applications across various regions and environments
- Built infrastructure-as-code with Pulumi, managing 100% of deployed resources with traceable, DRY code
- Automated developer tooling with Nix, allowing centralized management of developer environments that synchronized automatically between developer laptops and CI/CD
- Built incident response automation with incident.io, designing the tooling and processes for company-wide on-call rotation and participating in security and infrastructure shifts
- Automated observability for the company, unifying logs, metrics, and traces from applications and core infrastructure using Open Telemetry and Datadog
- Owned CI/CD pipelines, reducing runtimes by 30+ minutes while ensuring near-zero post merge failures

Grammatech, Inc — *Remote*

Scientific Software Engineer — *2019 - 2022*

- Optimized ML training times, reducing iteration speed from days to hours
- Designed and maintained CI/CD pipelines, drastically improving speed and facilitating cross-project integrity
- Built reusable developer environment tooling, enabling developers to rapidly switch between projects with disparate system requirements
- Designed framework and process for labeling large ground-truth dataset for ML training, allowing for automated detection and review of potentially mislabeled data
- Assessed academic codebase to patch assembly within precompiled binaries, generalizing development to be usable in industry settings and advising on future usage

Distributed Systems Lab, University of Pennsylvania — *Philadelphia, PA*

Senior Software Developer — *2017 - 2019*

- Headed the development of *DeDOS*, a ~20,000-line C framework for partitioning software across nodes to defend against DOS
- Benchmarked FPGAs and programmable network devices running at line-rate with microsecond-scale latency
- Discovered opportunities for program improvement through analysis of logs and network traces in python
- Automated techniques to rapidly and easily benchmark remote software for performance and functionality
- Designed *mysql* database schema for rapid insertion, viewing, and analysis of performance-monitoring statistics
- Built tool which injected code to gather and analyze network traffic, enabling real time anomaly detection of Node.JS and Apache servers
- Wrote up methods and evaluations for publication, creating clear and informative graphs demonstrating successful research
- Contributed to open-source behavioral model of SDN language (P4) to add periodically executed external functions
- Utilized *travis* and *check* to set up continuous integration which automatically unit-tests publicly pushed commits
- Administered and configured 10-node Linux cluster used for development and analysis of latency-critical applications
- Mentored and managed five undergraduate and master's student employees

Computational Memory Lab, University of Pennsylvania

Philadelphia, PA

Scientific Software Developer *2012-2017*

- Overhauled code in Python, C++, and MATLAB for real-time neural analysis and stimulation through human intracranial EEG
- Re-architected lab's data storage, creating quality controlled repository of searchable and analyzable neural and behavioral data
- Automated behavioral and neural data upload, analysis, and report generation workflow
- Designed software and interface using PyQT to locate electrodes in 3-D brain scans
- Created system for medical personnel at eight remote sites to easily run and update experiments
- Designed and ran experiment testing the effects of repetition on serial-order memory

High Energy Physics, University of Pennsylvania

Philadelphia, PA

Software Developer *2009-2013*

- Studied design of high-energy sensors to test low-level components, troubleshooting both analog and digital circuitry
- Designed user interface to automate physical testing and data analysis of circuit components
- Utilized hardware-descriptor language to automate readout of analog-to-digital converters in prototype proton-therapy device

Formal **Education**

Master of Science in Engineering: Computer & Information Science

Philadelphia, PA

University of Pennsylvania | GPA: 3.9 *2017-2019*

Thesis *Socket Launcher*: a framework for lossless, transparent, and rapid migration of TCP connections between servers

Coursework Software Systems, Networked Systems, Database and Information Systems, Big Data Analytics, Analysis of Algorithms, Introduction to Machine Learning, Theoretical and Computational Neuroscience, PhD Seminar on Network Research

Bachelor of Arts: Biological Basis of Behavior (Magna Cum Laude)

Philadelphia, PA

University of Pennsylvania | GPA: 3.67 *2012-2015*

Awards LPS Award for Academic Achievement in the Natural Science

Coursework Operating Systems, Software Engineering and Development, Mathematical Foundations of Computer Science

Public **Projects**

Article Friend

https://github.com/isaac-ped/aphasiaFriendly

Article friend is a tool for automating the generation of aphasia-friendly abstracts for scientific publications. You can view and use the tool live at *www.article-friend.com*, and read more about how we're improving the tool on our poster from the Society for the Neurobiology of Language 2026 Conference

DeDOS: Declarative Dispersion-Oriented Software

https://github.com/dedos-project/DeDOS

DeDOS aims to be a new software architecture which enables software to be run in a distributed manner across multiple servers. When a program is written in DeDOS's framework, communication between nodes is automatically handled, allowing the developer to program a series of function calls which are then dispersed among the resources in a data-center. With fine-grained software splitting, DeDOS can more fully utilize all available resources.

Socket Launcher

https://github.com/isaac-ped/Socket-Launcher

Socket Launcher is a framework for the lossless, transparent, and rapid migration of TCP connections between servers. Consisting of a proxy and a client library, Socket Launcher utilizes the extended Berkeley Packet Filter (eBPF) to perform rapid packet rewriting and retransmission, allowing for sub-millisecond user-space TCP transfer, and requiring no modifications to the linux kernel or TCP stack.

Academic **Publications**

What is "Aphasia-Friendly" About Aphasia-Friendly Materials? A Sandbox Series Presentation [*link*]

Deborah Levy, Anna Kasdan, Ryan Jensen, **Isaac Pedisich**

Society for the Neurobiology of Language Annual Meeting, 2026

A practical guide to translating scientific publications into aphasia-friendly summaries [*link*]

Anna V Kasdan, Deborah F Levy, **Isaac Pedisich**, Stephen M Wilson, Dominique Herrington

Perspectives of the ASHA special interest groups, 2025

The Penn Electrophysiology of Encoding and Retrieval Study [*link*]

Michael J Kahana, Lynn J Lohnas, M Karl Healey, Ada Aka, Adam W Broitman, Patrick Crutchley, Elizabeth Crutchley, Kylie H Alm, Brandon S Katerman, Nicole E Miller, Joel R Kuhn, Yuxuan Li, Nicole M Long, Jonathan Miller, Madison D Paron, Jesse K Pazdera, **Isaac Pedisich**, Joseph H Rudoler, Christoph T Weidemann

Journal of Experimental Psychology: Learning, Memory, and Cognition, 2024

When idling is ideal: Optimizing tail-latency for heavy-tailed datacenter workloads with perséphone [*link*]

Henri Maxime Demoulin, Joshua Fried, **Isaac Pedisich**, Marios Kogias, Boon Thau Loo, Linh Thi Xuan Phan, Irene Zhang

Proceedings of the ACM SIGOPS 28th Symposium on Operating Systems Principles, 2021

Theta-burst stimulation entrains frequency-specific oscillatory responses [*link*]

Ethan A Solomon, Michael R Sperling, Ashwini D Sharan, Paul A Wanda, Deborah F Levy, Anastasia Lyalenko, **Isaac Pedisich**, Daniel S Rizzuto, Michael J Kahana

Brain Stimulation, 2021

Flightplan: Dataplane disaggregation and placement for p4 programs [*link*]

Nik Sultana, Jon Sonchack, Hans Giesen, **Isaac Pedisich**, Zhaoyang Han, Nishanth Shyamkumar, Shivani Burad, André DeHon and Boon Thau Loo

18th USENIX Symposium on Networked Systems Design and Implementation (NSDI 21), 2021

TMC: Pay-as-you-Go Distributed Communication [*link*]

Henri Demoulin, Nikos Vasilakis, John Sonchack, **Isaac Pedisich**, Vincent Liu, Boon Thau Loo, Linh Thi Xuan Phan, et al.

Proceedings of the 3rd Asia-Pacific Workshop on Networking (APNet), 2019

Detecting asymmetric application-layer denial-of-service attacks in-flight with finelame [*link*]

Henri Maxime Demoulin, **Isaac Pedisich**, Nikos Vasilakis, Vincent Liu, Boon Thau Loo, Linh Thi Xuan Phan

USENIX Annual Technical Conference (ATC), 2019

DeDoS: Defusing DoS with Dispersion Oriented Software [*link*]

Henri Maxime Demoulin, Tavish Vaidya, **Isaac Pedisich**, Bob Dimaiolo, Jingyu Qian, Chirag Shah, Yuankai Zhang, et al.

Annual Computer Security Applications Conference (ACSAC), 2018

Automated Detection and Mitigation of Application-level Asymmetric DoS Attacks [*link*]

Henri Maxime Demoulin, **Isaac Pedisich**, Linh Thi Xuan Phan, and Boon Thau Loo.

ACM SIGCOMM Workshop on Self-Driving Networks (SDN), August 2018

Demonstration of the DeDoS Platform for Defusing Asymmetric DDoS Attacks in Data Centers [*link*]

Henri Maxime Demoulin, Tavish Vaidya, **Isaac Pedisich**, Nik Sultana, Jingyu Qian, Bowen Wang, Yuankai Zhang, et al

SIGCOMM demonstration, 2017 *(First prize for the ACM Student Research Competition*

Closed-loop stimulation of temporal cortex rescues functional networks and improves memory [*link*]

Youssef Ezzyat, Paul A. Wanda, Deborah F. Levy, Allison Kadel, Ada Aka, **Isaac Pedisich**, Michael R Sperling et al.

Nature Communications, 2018

Direct Brain Stimulation Modulates Encoding States and Memory Performance in Humans [*link*]

Youssef Ezzyat, James E. Kragel, John F. Burke, Deborah F. Levy, Anastasia Lyalenko, Paul Wanda, Logan O'Sullivan, Katherine B. Hurley, Stanislav Busygin, **Isaac Pedisich**, et al.

Current Biology, 2017

Proximity of Substantia Nigra Microstimulation to Putative GABAergic Neurons Predicts Modulation of Human Reinforcement Learning [*link*]

Ashwin G. Ramayya, **Isaac Pedisich**, Deborah Levy, Anastasia Lyalenko, Paul Wanda, Daniel Rizzuto, Gordon H. Baltuch and Michael J. Kahana

Frontiers in Human Neuroscience, 2017

Expectation Modulates Neural Representations of Valence throughout the Human Brain [*link*]

Ashwin G. Ramayya, **Isaac Pedisich**, Michael J. Kahana

NeuroImage, 2015